

# CSC566 Project Proposal

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As the global landscape of data grows in both quantity and complexity, we are reaching a point where the scale of information surpasses our capacity for effective processing and analysis. Especially with the advanced machine learning methods that we see today, the need for effective feature reduction is apparent. There are many popular dimensionality reduction techniques out there, all with their own strengths and weaknesses. Our project proposes to take an unbiased look at 5 different feature visualization techniques and compare them on 2 different high-dimensional datasets. Below are our five methods we set out to compare:

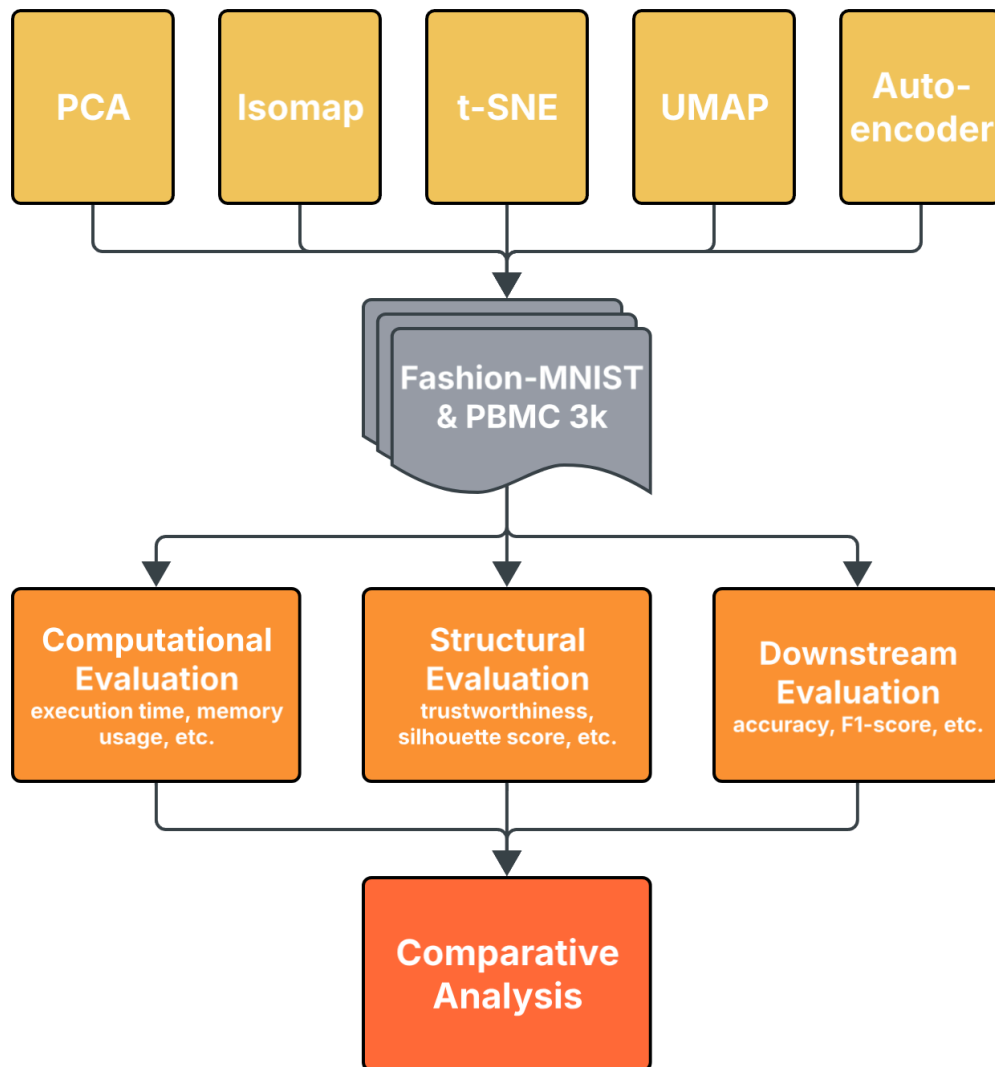
1. **PCA:** A classic linear dimensionality reduction technique
2. **UMAP:** A newer nonlinear dimensionality reduction technique focused on 2D visualizations of data clusters, built to analyze biological data
3. **t-SNE:** A slightly older nonlinear approach to visualizing data
4. **Autoencoders:** A neural network-based approach that can be adapted to visualize data
5. **Isomap:** Extension of MDS focused on visualizing data in 2D

We plan to implement our autoencoder fully from scratch, using Python and the PyTorch library. For the other methods, there exist popular Python libraries with efficient implementations that we can utilize for our testing. We will run all of our methods on each dataset, generating visualizations, as well as keeping track of metrics related to computational complexity. We will track things like memory usage, total computation time, and use this to compare each model. For the visualizations, we plan to not only perform a qualitative assessment, but also use metrics like the silhouette score, as well as the trustworthiness and continuity of the rankings. Beyond unsupervised clustering, we plan to evaluate each technique's ability to preserve predictive signals by training classification models on the reduced dimension representations. By comparing post reduction accuracy against the full feature baseline, we aim to determine which methods best maintain the underlying signal required for downstream machine learning tasks. The methodology is visualized in [Figure 1](#) below.

Below are the three datasets we plan to use:

- **Fashion-MNIST:** 70,000 samples with 784 features across 10 classes. This dataset offers a more complex manifold than standard digits to test if non-linear methods can better preserve subtle textures for classification.

- **PBMC 3k (scRNA-seq):** 2,700 samples with ~32,000 features and 8+ cell-type classes. This dataset's extreme sparsity and high dimensionality provide a rigorous test for identifying latent biological structures and maintaining predictive signals in noisy data.
- **Swiss roll / High-Dimensional Moons dataset:** This is a backup which is a dataset which we can generate. This not only allows us to ensure sufficient data size, but we can also control several hyperparams to change the shape of our dataset to see its impact on dimensionality reduction



**Figure 1: Methodology Diagram**